

# Four-Armed Chef: Designing Bodily Play with Agentic Supernumerary Limbs in Virtual Reality

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## Abstract

Supernumerary Robotic Limbs (SRLs) extend the human body beyond its natural limits. While prior work has primarily emphasized functional efficiency in industrial or medical contexts, the experiential potential of SRLs remains underexplored. In this work, we explore the design of bodily play with agentic supernumerary limbs. We present *Four-Armed Chef*, a VR cooking game in which a player embodies an avatar with two natural arms and two supernumerary arms. Controlled by a Reinforcement Learning agent, these limbs autonomously coordinate with the player to complete collaborative cooking tasks. Through preliminary first-person exploration, we iteratively refined the system's movement dynamics and visual representation to mitigate bodily conflicts. Our work offers an initial exploration of designing agentic SRLs for playful bodily interaction, establishing a foundation for future studies on experiential human-AI bodily integration.

## CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Interaction paradigms**; *Interaction design*.

## Keywords

Bodily Play, Supernumerary Robotic Limbs, Virtual Reality, Human-AI Collaboration

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## 1 Introduction

*Supernumerary Robotic Limbs (SRLs)* extend the human body schema by attaching extra arms, fingers, or other appendages [14, 23, 32, 36]. By providing additional limbs, SRLs expand human physical capabilities, enabling humans to perform beyond the limitations of their biological bodies. Building on this capability, prior work has explored SRLs in domains such as industrial assembly and medical rehabilitation [23–25], in which users can leverage SRLs to assist with tasks such as stabilizing objects, performing parallel actions, and supporting guided motor rehabilitation and training.

While most prior research has prioritized efficiency and task performance with concern for control algorithms and wearable interfaces [27, 28, 34], little attention has been paid to the experiential aspects of inhabiting a new body schema introduced by SRLs. Attending to experiential aspects is essential in the context of body schema extension, as such extensions fundamentally alter how users perceive, control, and relate to their own bodies. Play serves as an important lens for us to explore the experiential perspective, not merely as a frivolous activity, but as an approach that lowers barriers to adoption, encourages experimentation without fear of failure, and foregrounds the experiential qualities of technology rather than solely its utility [3, 4].

To explore this potential, we presented a VR bodily game called *Four-Armed Chef*, a VR system where a player embodies an avatar with four arms: two mapped to their physical movements and two autonomous limbs controlled by an AI agent. We set up a simple cooking task in which a human user and AI must coordinate their motions to complete recipes, navigating the tension between smooth cooperation and inevitable spatial conflicts. By situating SRLs in a virtual environment, we circumvent the hardware constraints of weight, fatigue, and safety, creating a low-stakes testbed for investigating human-AI co-embodiment. To support adaptive and egalitarian coordination between the player and agent, we employed reinforcement learning to drive the agent to control the SRLs, which allows the AI to automatically respond to human motion intentions, facilitating more fluid and context-aware collaboration [6, 7, 32, 37, 38]. We conducted a preliminary first-person exploration to probe initial experience with our supernumerary

AI arms, iteratively experimenting and refining the design during engagement with the system.

Our work makes the following contributions. Practically, we present a VR game *Four-Armed Chef*, specifically designed to explore playful experience with the human-AI integration of SRLs. We conducted a preliminary first-person exploration to engage with the supernumerary AI arms system and iteratively refine its design based on firsthand experience. We believe our work offers a method for designing supernumerary AI arms in playful contexts, offering a foundation for future studies to investigate bodily play, human-AI coordination, and the experiential potential of extended body schema.

## 2 Related Work

In this section, we reviewed prior works in supernumerary robotic limbs and human-AI co-embodiment, as well as experience bodily extension as play.

### 2.1 Supernumerary Robotic Limbs and Human-AI Co-embodiment

Supernumerary robotic limbs refer to wearable devices that can be attached to the human body to perform tasks [15, 32]. Unlike traditional wearable devices such as exoskeletons and prostheses that replace or augment existing limbs [30, 35], SRLs introduce additional, independently controlled limbs that extend human capability and support bodily coordination [23–25]. Beyond support, SRLs can also enhance human strength and precision [2, 5]. By introducing additional limbs, SRLs also give rise to novel forms of embodied experience, in which multiple limbs must be coordinated and integrated within a single body. For example, Muehlhaus et al. [20] deployed an additional wearable robotic arm and invited users to engage in open-ended bodily exploration, eliciting how people naturally move, coordinate, and interact with a supernumerary limb. Hongyu et al. [40] explored how users interact with two virtual supernumerary arms and investigated the control strategies users adopt to appropriate supernumerary arms and facilitate coordination. This experiential challenge resonates with research on virtual co-embodiment, which demonstrates that humans can adopt a strange body shared with another individual, thereby foster playful experiences and effectively support the learning of physical skills [8, 9, 11, 13, 16, 26]. For example, Saraiji et al. [29] developed a system aimed to support remote collaboration where the extended arms were controlled by another person, creating a sense of bodily collaboration across distance. Similarly, body-part-segmented virtual co-embodiment techniques allow two users to inhabit a single avatar while independently controlling different limbs [10, 18, 39]. Together, these approaches suggest that the human sensorimotor system can accommodate distributed control across multiple limbs, an insight that is directly relevant to the design and experience of supernumerary robotic limbs.

In this work, we focus on the experiential aspect of inhabiting a body attached with supernumerary robotic limbs. We examine how users experience, coordinate, and make sense of supernumerary robotic limbs as part of their body. To support this investigation, we draw inspiration from prior research on virtual co-embodiment

and design a setting in which a human user and an AI agent are jointly embodied within a single virtual body.

### 2.2 Experience Bodily Extension as Play

Body extension refers to the use of interactive technologies to extend the body schema, such as wearables and computational devices worn on or attached to the body, facilitating users' bodily capabilities. [19, 28, 33]. Rather than merely extending what the body can do, these technologies can also reshape how the body is experienced. Recent studies have increasingly drawn attention to the experiential qualities of body extension. For example, Nishida et al. [22] developed HandMorph, a passive hand exoskeleton that miniaturizes grasp, allowing users to experience the sensation of having a child-sized hand. By foregrounding the experiential dimension of bodily extension, this work provided valuable insights into the design of children's toys. Beyond enhancing what the body can do, such technologies can also transform how the body is experienced, opening up opportunities for playful engagement with one's own body [21]. Building on this, bodily extension can also be framed as playfulness. Svanæs et al. [31] investigated the integration of wearable tails into the body schema and proposed the concept of kinesthetic body extension. Their study demonstrated that such bodily extensions can evoke novel and humorous experiences, extending the role of body extension into entertainment and play contexts.

Inspired by these works, we want to explore the potential of designing SRLs to support playful experiences as a form of bodily extension. We believe that SRLs have the potential to offer playfulness to players in bodily game, allowing players to explore and experiment in extended body schema, engage in novel bodily coordination, and experience their extended body in playful and imaginative ways.

## 3 Four-Armed Chef

In this work, we proposed *Four-Armed Chef*, a VR game featuring a four-armed avatar that two natural arms were controlled by the player and two supernumerary arms controlled by an AI agent [2, 23, 25]. The goal of this work is to explore the user experience of Human-AI collaboration under a virtual co-embodiment paradigm. To this end, we designed a cooperative cooking task that requires tight coordination between the player and the AI agent. The human and the AI must collaboratively control the avatar's four arms to process food items, introducing challenges related to shared task awareness, action coordination, and potential motion conflicts between human- and AI-controlled limbs.

### 3.1 Game Design

*Four-armed Chef* is a VR body game in which a player and an AI agent co-embodiment a four-armed avatar, coordinating to finish a cooking task (see Fig. 1). The avatar is designed as a natural human body augmented with two supernumerary arms attached at the shoulders. The player's real-world body movements are captured in real time and mapped onto the avatar's natural body parts, while the AI agent controls the supernumerary arms to coordinate with the human player to accomplish the game task. During the game, the AI agent dynamically monitors the state of the environment and

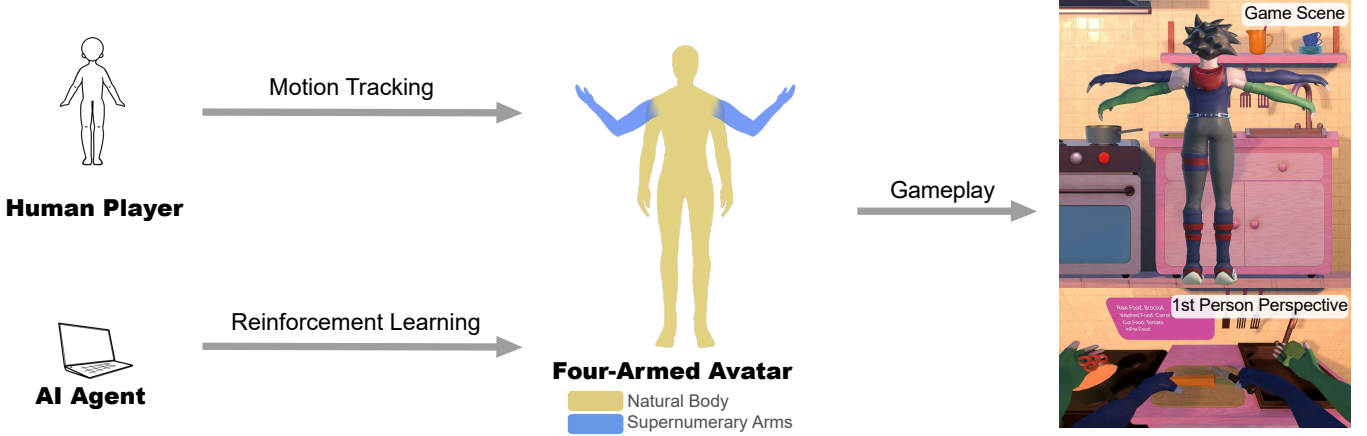


Figure 1: System Diagram of *Four-Armed Chef*.

the movements of the natural arms, leveraging these information to guide the supernumerary AI arms in performing complementary actions as well as avoiding potential conflicts. Through this shared embodiment, both agents jointly manipulate the environment and negotiate control over the avatar’s four limbs.

**Cooking Task Design** The game is set in a virtual kitchen where three food items need to be prepared: a tomato, a carrot, and a broccoli. Each item requires a fixed sequence of processing steps, i.e., washing in the sink, chopping on a cutting board, and finally transferring into a cooking pot. A canvas-based UI panel presents the current processing state of each item, enabling both the human player and the AI agent to monitor task progression and strategically plan their subsequent actions. The game task is completed once all food items have been fully processed.

The task is intentionally designed to support egalitarian collaboration between the human and the AI, rather than a hierarchical or assistant-based relationship. In our game, the recipe involves multiple steps, while none are explicitly assigned to either the player or the agent. Instead, the player and the agent must organically coordinate to divide the labor. Operating concurrently, the human and the AI-controlled supernumerary arms continuously adapt to each other’s actions and the evolving state of the kitchen. For instance, while the human player is chopping a food item, the AI can simultaneously handle complementary steps, such as washing other items, demonstrating fluid, parallel cooperation.

## 3.2 Implementation

In this section, we present the details of system implementation.

**3.2.1 VR Setup.** We developed *Four-Armed Chef* using the Unity3D platform. For the virtual reality experience, we utilized the HTC VIVE Pro2 Head-Mounted Displays (HMD)<sup>1</sup>, which enables players to fully immerse themselves in the virtual environment and embody their presence within the avatar. Each HMD provides a pixel resolution of  $2448 \times 2448$  per eye, a refresh rate of 120 Hz, and a

field of view that spans 120 degrees. We used two controllers to track the movements of users.

**3.2.2 Human-AI Co-Embodiment.** Central to our system is the seamless integration of human and AI control within a single avatar. We employed the Azure Kinect DK<sup>2</sup> to capture the player’s real-time motion and map it to the natural body parts of the virtual avatar. Simultaneously, the trained AI agent was embedded into two supernumerary arms. These arms are attached to the corresponding left and right shoulder joints of the avatar. This co-embodiment architecture ensures that both the human-controlled and AI-controlled arms share the same physical reference frame, enabling spatial awareness and shared manipulation of the environment.

**3.2.3 Supernumerary AI Arms Control.** To generate coordinated motion for the supernumerary arms, we first explored inverse kinematics (IK) [1]. With this approach, the movements of the supernumerary arms were expressed as target-reaching behaviors. However, our preliminary tests suggested that IK-based control lacked adaptability to the player’s unpredictable movements, frequently resulting in unrecognized intersections and bodily conflicts. This forced the human player to actively avoid the AI, violating our core design goal of egalitarian collaboration.

To address this limitation, we implemented a reinforcement learning (RL) agent [6, 32, 37]. Unlike traditional rigid control that only supports pre-scripted or independent action, RL models human-AI coordination as an interactive process where the agent continuously evolves its actions based on the human’s ongoing movement within the task context, allowing implicit negotiation of shared workspace and task responsibilities through interaction.

To operationalize this co-adaptive human-AI coordination, we formulated the problem in an RL framework, specifying the agent’s observations, actions, and reward signal.

- **Observations:** The agent perceives the shared workspace by simultaneously tracking the status of task-relevant items and the real-time kinematics of the player’s natural arms. By

<sup>1</sup><https://business.vive.com/hk/product/vive-pro2/>

<sup>2</sup><https://azure.microsoft.com/en-us/products/kinect-dk>

understanding both the current task state and the human's ongoing movements, the agent can infer user player intent, anticipate potential conflicts and thus make decisions about where and how to intervene.

- **Actions:** The agent outputs continuous control signals for the supernumerary arms, adaptively modulating joint rotations and actuation strengths at each time step to drive their movements. This allows the AI to fluidly adjust its embodied configuration and complement the human's movements in real time.
- **Rewards:** The reward function defines the learning objective by jointly encouraging task completion and safe physical coordination. The agent receives a positive reward when an AI-controlled hand successfully reaches the target position required for the current culinary step, incentivizing purposeful, task-oriented motion. Conversely, the agent incurs negative penalties when the supernumerary arms collide with the player's natural arms, with each other, or with unintended body parts, discouraging disruptive behaviors and promoting harmonious co-use of the shared workspace.

We trained the RL agent using Unity ML-Agents [12], an open-source toolkit which enables interactive simulations to be used directly as training environments for intelligent agents. Training was performed using Proximal Policy Optimization (PPO) for a total of 30 million environment steps. The policy and value networks each consisted of three fully connected layers with 256 hidden units and received normalized observations. We used a batch size of 2048 and a buffer size of 20480, with a time horizon of 1000 steps. The learning rate was set to  $3 \times 10^{-4}$  with a linear decay schedule, and the discount factor and GAE parameter were  $\gamma = 0.995$  and  $\lambda = 0.95$ , respectively.

#### 4 Preliminary First-Person Exploration

The system design evolved through an iterative process. During our design process, we conducted a first-person exploration to evaluate the prototype [17]. Here, the first author personally experienced the game and identified discomfort and interaction breakdowns, then the author team discussed these issues and iteratively refined the system design. This process highlighted three critical areas for improvement that directly informed the current design.

First, we adjusted the motion behavior of supernumerary AI arms. Regarding movement dynamics, we observed that although the arms could accurately identify targets and act accordingly, their initial velocity and rotation speeds were excessively fast. This kinematic mismatch with natural human movement, broke immersion and created a sense of bodily interference, making it difficult to coordinate the natural arms with the rapid motions of supernumerary AI arms. To address this, we dampened the AI-controlled arms to act at a speed consistent with human motor capabilities, providing the player sufficient time to react and coordinate smoothly with the supernumerary limbs.

Second, we modified how the agent anticipates and avoids potential conflicts between the natural arms and supernumerary AI arms. In the initial setup, all four arms were equipped with physical colliders, and conflicts were detected based on whether their colliders came into contact. When a collision was detected, the

supernumerary AI arms were pushed away through the physics engine's collision response, after which the agent re-planned their movements. However, this often resulted in repeated or prolonged collisions, leading to a frustrating sense of miscoordination. To mitigate this, the RL agent was designed to anticipate potential collisions. When the supernumerary AI arms' trajectories approached the natural arms, the agent proactively issued commands to steer them away and re-plan their movements, preventing collisions before they occurred.

Third, we refined the task complexity. The initial prototype included elaborate steps, such as retrieving utensils from a cabinet. However, we found this procedural complexity monopolized the player's attention, distracting them from the core experience of perceiving and coordinating with the supernumerary arms. To address this, we iterated the design, simplifying the task to focus solely on processing the three food items. This allows players to more easily adopt and explore the supernumerary AI arms without being burdened by task complexity.

Finally, we adjusted the visual representation to resolve ambiguity. Initially, all four arms were rendered with an identical appearance to promote a unified sense of embodiment. However, we found that this caused confusion between the natural arms and the AI-controlled arms, making it difficult to accurately localize and coordinate their own movements. To address this ambiguity, the supernumerary AI arms are rendered with a distinct visual appearance from the natural arms. This creates a clear contrast with the user's own arms, enabling participants to easily distinguish between the movements of the self and the AI agent.

#### 5 Future Work

We designed and iterated a VR bodily game as a first step toward exploring how supernumerary AI arms are experienced. In future work, we plan to extend this investigation in several directions. First, we will recruit participants for a formal user study through both qualitative and quantitative methods. Through this study, we will evaluate the overall performance and generalizability of the system across users. We will also examine how players perceive and adopt supernumerary AI arms, investigate how they coordinate with them, and explore how the extended body schema influences their bodily experience. We aim to further uncover the potential of supernumerary AI arms in supporting both novel forms of bodily engagement and playful experiences. Second, we will explore additional scenarios beyond the cooking task, investigating how supernumerary AI arms shape embodiment and interaction across diverse contexts. Finally, we will further investigate the potential of supernumerary AI arms as a medium for studying bodily experience, examining how they can contribute to understanding embodiment, human-AI integration.

#### 6 Conclusion

In this work, we explored the design of playful experience with supernumerary AI arms. We presented *Four-Armed Chef*, a VR game in which a player embodies a single avatar with two supernumerary arms controlled by an AI agent. Through a preliminary first-person exploration, we engaged with our supernumerary AI arms

system to iteratively refine the design and investigate initial interactions. Ultimately, we hope our work expands the possibilities for designing SRLs at an experimental level, not only enhancing human adaptation to supernumerary limbs but also advancing human–AI integration and collaboration.

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